DATA

DATA MINING

BSCS 3A

CHRISTIAN SY

DATA

- Attributes/Features
- Types of Data Sets
- Data Quality
- **Data Preprocessing**
- Similarity and Dissimilarity
- Density

DATA PREPROCESSING

Data preprocessing is the process of **transforming raw data into a clean, structured, and suitable format** for analysis or modeling.

It improves:

- Accuracy
- Efficiency
- Model performance
- Interpretability

Data Preprocessing Methods:

- Aggregation
- Sampling
- Dimensionality Reduction
- Feature subset selection
- Feature creation
- Discretization and Binarization
- Attribute Transformation

AGGREGATION

- Aggregation is the process of combining multiple records or attributes into a summarized form to reduce data size and improve analysis. It reduces noise, increases data stability, and transforms data into higher-level information.
- Purpose

  Data reduction
  - Reduce the number of attributes or objects

  Change of scale
  - Cities aggregated into regions, states, countries, etc

  More “stable” data
  - Aggregated data tends to have less variability (e.g., reduce seasonality by aggregation to yearly data)

Date	Sales
Jan 1	1,000
Jan 2	1,200
Jan 3	900

Month	Total Sales
January	3,100

AGGREGATION

Real-World Applications

- ✓ **Retail analytics** – monthly revenue per branch
- ✓ **Banking** – total deposits per quarter
- ✓ **Disaster resilience research (like your microenterprise project)** – total business losses per municipality
- ✓ **IoT systems** – hourly average sensor readings

Why Important in Data Mining?

- ✓ Reduces dataset size
- ✓ Improves computational efficiency
- ✓ Highlights trends
- ✓ Reduces variance and noise

SAMPLING

- Sampling is the main technique employed for data selection.

  It is often used for both the preliminary investigation of the data and the final data analysis.
- Statisticians sample because **obtaining** the entire set of data of interest is too expensive or time consuming.
- Sampling is used in data mining because **processing** the entire set of data of interest is too expensive (e.g., does not fit into memory or is too slow).

Real-World Example

- Retail: Millions of sales transactions → sample 10,000 to find frequent itemsets
- Social media: Streaming data → sample tweets for sentiment analysis
- Healthcare: Sample patient records to detect disease patterns

SAMPLING …

- The key principle for effective sampling is the following:
  - Using a sample will work almost as well as using the entire data sets, if the sample is **representative**.
  - A sample is representative if it has approximately the same property (of interest) as the original set of data.

TYPES OF SAMPLING ALONG REPLACEMENT AND SELECTION DIMENSIONS

Replacement?

Sampling Without Replacement
Each selected item is **removed from the population**, so it **cannot be selected again**.
The population size decreases after each draw.

- ✓ No duplicates in the sample
- ✓ Probability changes after each selection

Example: Drawing cards from a deck and not putting them back.

Sampling With Replacement
Selected items are **returned to the population** before the next draw.
The population size remains the same for every selection.

- ✓ Duplicates are possible
- ✓ Probability stays constant for each draw

Example: Rolling a die multiple times.

TYPES OF SAMPLING ALONG REPLACEMENT AND SELECTION DIMENSIONS

Selection?

Simple Random Sampling (SRS)
Every member of the population has an **equal probability of being selected**.
Additionally, **every possible sample of size n has an equal chance of being chosen**.

- ✓ Selection is purely random
- ✓ No grouping before sampling
- ✓ Can be with or without replacement

Example: Randomly selecting 50 student IDs from the entire university list.

Stratified Sampling
The population is first **divided into homogeneous subgroups (strata)** based on a characteristic.
Then, **random samples are drawn from each stratum**.

- ✓ Ensures representation from all groups
- ✓ Reduces sampling bias
- ✓ Often improves estimation accuracy

Example, If a university has:

- ➢ 60% freshmen
- ➢ 40% seniors

You sample proportionally:

- ➢ 60% from freshmen
- ➢ 40% from seniors

SAMPLE SIZE DETERMINATION

Determining **sample size** is a critical step in data mining, statistics, and machine learning because it affects:

- ✓ Model accuracy
- ✓ Statistical validity
- ✓ Computational cost
- ✓ Generalizability of results

WHY SAMPLE SIZE MATTERS

If the sample is:

- ✓ **Too small** → unreliable results, high variance, overfitting
- ✓ **Too large** → unnecessary computational cost
- ✓ **Biased** → invalid conclusions regardless of size

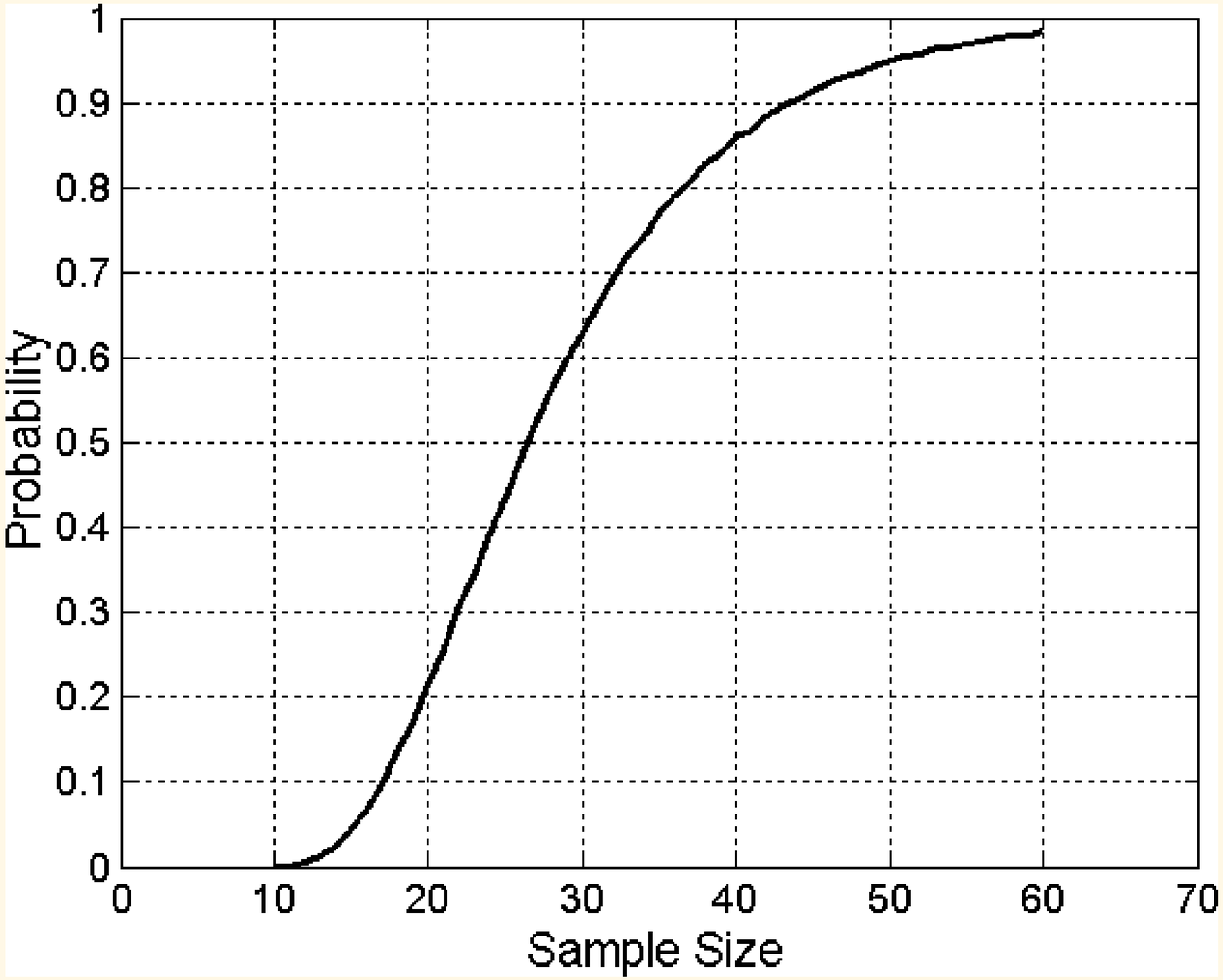
In data mining, sample size affects:

- ✓ Classification accuracy
- ✓ Confidence in pattern discovery
- ✓ Stability of clustering
- ✓ Reliability of association rules

SAMPLE SIZE DETERMINATION

- What sample size is necessary to get at least one object from each of 10 groups.

  - Sample size determination:
    1. Statistical inference
    2. Machine learning modeling
    3. Data mining pattern discovery



1. SAMPLE SIZE DETERMINATION IN STATISTICAL INFERENCE

In **statistical inference**, sample size is determined to ensure that estimates of population parameters (mean, proportion) are:

- ✓ Accurate
- ✓ Reliable
- ✓ Within a specified margin of error
- ✓ At a chosen confidence level

The goal is **generalization from sample to population**.

Key Principle

Sample size depends on:

1. Confidence level (e.g., 95%)
2. Margin of error
3. Population variability
4. Population size (if small)

1. SAMPLE SIZE DETERMINATION IN STATISTICAL INFERENCE

Formula-Based Determination

For estimating a mean:

$$n = \frac{Z^2\sigma^2}{E^2}$$

For estimating a proportion:

$$n = \frac{Z^2p(1-p)}{E^2}$$

Suppose you want to estimate:

- ✓ The average income of microenterprises in a province
- ✓ With 95% confidence
- ✓ Margin of error ±5%

You compute required sample size using statistical formulas.

SAMPLE SIZE IN STATISTICS - ESTIMATING A MEAN

Example: Average Monthly Income of Microenterprises

Problem

You want to estimate the **average monthly income of microenterprises in a province** with:

- ✓ 95% confidence
- ✓ Margin of error = ±5%
- ✓ From previous studies, standard deviation (σ) = ₱15,000
- ✓ Estimated mean income ≈ ₱60,000

Common Z Values

Confidence Level	Z Value
90%	1.645
95%	1.96
99%	2.576

Example: Average Monthly Income of Microenterprises

Step 1: Convert Margin of Error to Actual Value

5% of ₱60,000:

$$E = 0.05 \times 60{,}000 = 3{,}000$$

So:

$$E = ₱3{,}000$$

Step 2: Use Sample Size Formula for Mean

$$n = \frac{Z^2\sigma^2}{E^2}$$

For 95% confidence:

$$Z = 1.96$$

Substitute values:

$$n = \frac{(1.96)^2(15000)^2,}{(3000)^2,}$$

Step 3: Compute

$$(1.96)^2 = 3.8416$$

$$(15{,}000)^2 = 225000000$$

$$(3{,}000)^2 = 9000000$$

Now:

$$n = \frac{3.8416 \times 225{,}000{,}000}{9{,}000{,}000}$$

$$n = \frac{864{,}360{,}000}{9{,}000{,}000}$$

$$n \approx 96.04$$

Final Answer

You need **at least 97 microenterprises**.

Interpretation

- ✓ If you survey 97 microenterprises:
- ✓ You can estimate the average income within ±₱3,000 with 95% confidence.

SAMPLE SIZE IN STATISTICS - ESTIMATING A PROPORTION

Example: Average Monthly Income of Microenterprises

Problem

You want to estimate:

- ✓ Percentage of microenterprises that are profitable
- ✓ 95% confidence
- ✓ Margin of error = ±5%
- ✓ No prior estimate of proportion → use p = 0.5 (most conservative)

SAMPLE SIZE IN STATISTICS - ESTIMATING A PROPORTION

Step 1: Use Formula for Proportion

$$n = \frac{Z^2 p(1-p)}{E^2}$$

Where:

Z = 1.96
p = 0.5
E = 0.05 (margin of error ±5)

Step 2: Substitute Values

$$n = \frac{(1.96)^2(0.5)(0.5)}{(0.05)^2}$$

Step 3: Compute

$$(1.96)^2 = 3.8416$$
$$0.5 \times 0.5 = 0.25$$
$$(0.05)^2 = 0.0025$$

Now:

$$n = \frac{3.8416 \times 0.25}{0.0025}$$
$$n = \frac{0.9604}{0.0025}$$
$$n = 384.16$$

Final Answer

You need **at least 385 microenterprises**.

Interpretation

If 385 are sampled:
You can estimate the true proportion of profitable microenterprises within ±5% with 95% confidence.

SAMPLE SIZE IN STATISTICS - MEAN VS PROPORTION

Type	Required Sample	Why Larger?
Mean	97	Depends on standard deviation
Proportion	385	Maximum variability assumed (p=0.5)

Proportion often requires larger samples.

1. For mean → need estimate of standard deviation.
2. For proportion → if p unknown, use 0.5.
3. Smaller margin of error → much larger sample.
4. Higher confidence → larger sample.

2. SAMPLE SIZE DETERMINATION IN MACHINE LEARNING MODELING

In machine learning, sample size is determined to:

- ✓ Train predictive models
- ✓ Minimize overfitting
- ✓ Improve generalization
- ✓ Achieve stable performance

Unlike statistical inference, the focus is **prediction accuracy**, not parameter estimation.

Key Principle

More data usually improves performance — but only up to a point.

Instead of formulas, we use:

- ✓ Learning curves
- ✓ Cross-validation
- ✓ Empirical evaluation

2. SAMPLE SIZE DETERMINATION IN MACHINE LEARNING MODELING

A. Learning Curve Analysis

Procedure:

- ✓ Train model with increasing dataset sizes.
- ✓ Plot training and validation accuracy.
- ✓ Identify plateau (performance stabilizes).
- ✓ If performance stops improving → sufficient data.

B. Rule-of-Thumb Guidelines

- ✓ Logistic regression: at least 10–20 samples per feature
- ✓ Decision trees: hundreds to thousands
- ✓ Deep learning: thousands to millions
- ✓ Rare event prediction: large dataset needed

Example

Suppose:

- ✓ 20 features predicting business failure.
- ✓ Rule: 20 samples per feature.

Minimum sample size ≈ 400 observations.

3. SAMPLE SIZE DETERMINATION IN DATA MINING PATTERN DISCOVERY

In pattern discovery (e.g., clustering, association rules), sample size determines:

- ✓ Stability of discovered patterns
- ✓ Reliability of associations
- ✓ Meaningfulness of clusters

The goal is **discovering hidden structures**, not prediction or estimation.

A. Association Rule Mining

In market basket analysis:

- ✓ Larger sample → more reliable support values
- ✓ Rare itemsets require large datasets

Example:

To discover pattern: “Flood → Business closure”

If flood events are rare, you need many observations to detect reliable patterns.

3. SAMPLE SIZE DETERMINATION IN DATA MINING PATTERN DISCOVERY

B. Clustering

Clusters become stable only when:

- ✓ Sample size captures full variability
- ✓ Different subgroups are sufficiently represented

Small sample → unstable clusters

C. Anomaly Detection

Rare events require large datasets.

Example:

Fraud detection with 0.5% fraud rate:

- ✓ Need large sample to detect fraud patterns reliably.

3. SAMPLE SIZE DETERMINATION IN DATA MINING PATTERN DISCOVERY

Methods for Determining Sample Size in Pattern Discovery

1. Stability testing
   - ✓ Run clustering on different samples
   - ✓ Check if clusters remain consistent
2. Support threshold analysis
   - ✓ Ensure sufficient minimum support
3. Subsampling validation
   - ✓ Compare patterns across subsets

3. SAMPLE SIZE DETERMINATION IN DATA MINING PATTERN DISCOVERY

To summarize

Context	Goal	How Sample Size is Determined	Nature
Statistical Inference	Estimate population parameter	Mathematical formulas	Theoretical
Machine Learning	Predict accurately	Learning curves, validation	Empirical
Pattern Discovery	Discover hidden structures	Stability testing, support levels	Exploratory

DIMENSIONALITY REDUCTION (FEATURE EXTRACTION)

In data mining and machine learning, **dimensionality** refers to the number of features (variables, attributes, predictors) in a dataset.

Example:

Business_ID	Sales	Profit	Employees	Location	Years_Operation	Social_Media_Score

Here, the dataset has **6 dimensions (features)**.
When datasets contain:

- ✓ 50 features → moderate dimensionality
- ✓ 500 features → high dimensionality
- ✓ 10,000+ features → very high dimensionality

High dimensional data is common in:

- ✓ Text mining (each word = feature)
- ✓ Image processing (each pixel = feature)
- ✓ Genomics (each gene = feature)
- ✓ IoT sensor systems (multiple sensor readings)

DIMENSIONALITY REDUCTION (FEATURE EXTRACTION)

Dimensionality Reduction is the process of reducing the number of input variables while preserving important information.

It helps:

- ✓ Improve model performance
- ✓ Reduce computation time
- ✓ Prevent overfitting
- ✓ Improve visualization

THERE ARE TWO MAIN APPROACHES:

A. Feature Selection

Select a subset of existing features.

Example:

From 50 business indicators → select only 10 most predictive features.

Methods:

- ✓ Correlation filtering
- ✓ Chi-square test
- ✓ Mutual information
- ✓ Recursive Feature Elimination (RFE)

THERE ARE TWO MAIN APPROACHES:

B. Feature Extraction

Create new features from existing ones.

Instead of selecting features, we transform them into a smaller set of new features.

Example:

✓ From 100 correlated financial indicators → reduce to 5 principal components.

COMMON DIMENSIONALITY REDUCTION TECHNIQUES

1. Principal Component Analysis (PCA) is an **unsupervised linear dimensionality reduction technique** that transforms correlated variables into a smaller number of uncorrelated variables called **principal components**. It preserves **maximum variance**.

Used in:

- ✓ Pattern recognition
- ✓ Image compression
- ✓ Financial risk modeling

Example (Microenterprise dataset):

- ✓ Sales
- ✓ Expenses
- ✓ Revenue growth
- ✓ Profit margin
- ✓ Cash flow
- ✓ Inventory turnover

PCA might reduce these 6 correlated financial indicators into:

Component 1: Overall Financial Strength
Component 2: Growth Trend

These components capture most of the variance.

COMMON DIMENSIONALITY REDUCTION TECHNIQUES

2. Linear Discriminant Analysis (LDA)

Used for supervised dimensionality reduction.

Maximizes class separability.

Example:

Classifying (microenterprises into):

- ✓ Resilient
- ✓ Moderately Vulnerable
- ✓ Highly Vulnerable

LDA reduces many indicators into fewer features that best separate these categories.

COMMON DIMENSIONALITY REDUCTION TECHNIQUES

3. t-SNE (t-Distributed Stochastic Neighbor Embedding)

✓ Used for visualization of high-dimensional data.
✓ Common in image and deep learning datasets.

4. Autoencoders (Deep Learning)

Neural networks that learn compressed representations.
Used in:

✓ Anomaly detection
✓ Image compression
✓ Feature learning

WHY HIGH-DIMENSIONAL DATA IS PROBLEMATIC

High-dimensional data leads to three major issues:

- ✓ Curse of Dimensionality
- ✓ Overfitting
- ✓ Increased Computational Cost

CURSE OF DIMENSIONALITY

The term **Curse of Dimensionality** was introduced by **Richard Bellman** in the context of dynamic programming.

In data mining, it refers to:

The set of problems that arise when analyzing and organizing data in high-dimensional feature spaces.

As the number of attributes (dimensions) increases:

- ✓ Data becomes sparse
- ✓ Distance measures lose discriminative power
- ✓ Computational cost grows rapidly
- ✓ Model performance may degrade

CURSE OF DIMENSIONALITY

Imagine:

- 1 dimension → a line
- 2 dimensions → a square
- 3 dimensions → a cube

As dimensions increase:

The volume of the space grows exponentially.

If you want to maintain the same data density:

- 10 points in 1D
- 100 in 2D
- 1,000 in 3D
- 1,000,000 in 6D

Data requirements explode exponentially.

CURSE OF DIMENSIONALITY

Distance Becomes Meaningless

Many algorithms rely on distance:

- ✓ K-Nearest Neighbors (KNN)
- ✓ K-Means clustering

In high dimensions:

- ✓ The difference between nearest and farthest neighbor shrinks.
- ✓ Everything becomes almost equally distant.

This weakens similarity-based learning.

CURSE OF DIMENSIONALITY

Real Example: Text Mining

In a document classification task:

- 10,000 unique words
- Each word = 1 feature

Most documents contain only a few words → extremely sparse matrix.

This creates:

- Poor clustering
- Unstable classification boundaries

Dimensionality reduction like PCA or embeddings solves this.

OVERFITTING

Overfitting occurs when a model:

- Learns noise instead of patterns
- Performs very well on training data
- Performs poorly on new/unseen data
- High dimensionality increases risk.

If:
Number of features ≈ Number of samples
The model can memorize the dataset.
Example:

- 100 samples with 95 features
- A decision tree or neural network can perfectly fit training data — but generalization fails.

INCREASED COMPUTATIONAL COST

More features → more calculations → slower algorithms.

Why It Happens

Many algorithms depend on number of features:

- ✓ Distance calculations in KNN → $O(n \times d)$
- ✓ Matrix operations in PCA → $O(d^2)$
- ✓ Neural network parameters increase with feature size

Where:

- ✓ n = number of samples
- ✓ d = number of features

If d is large → computation becomes expensive.

HIGH-DIMENSIONAL DATA

Problem	What Happens	Why It's Bad
Curse of Dimensionality	Data becomes sparse	Distance loses meaning
Overfitting	Model memorizes noise	Poor generalization
Computational Cost	Slower training	Higher resource use

Key Takeaways

1. More features ≠ Better model
2. High-dimensional data requires exponentially more samples
3. Dimensionality reduction improves:
   - Accuracy
   - Speed
   - Interpretability
   - Scalability
4. Always analyze feature importance before modeling

DATA

- Attributes/Features
- Types of Data Sets
- Data Quality
- Data Preprocessing
- **Similarity and Dissimilarity**
- Density

SIMILARITY AND DISSIMILARITY

In data mining, **similarity** measures how alike two data objects are, while **dissimilarity** measures how different they are.

- ✓ **High similarity** → objects are close or alike
- ✓ **High dissimilarity** → objects are far apart or different

Most machine learning and data mining algorithms depend on these concepts.

SIMILARITY AND DISSIMILARITY

Similarity and dissimilarity are used in:

- ✓ Clustering (K-Means, Hierarchical Clustering)
- ✓ Classification (K-Nearest Neighbors)
- ✓ Recommendation Systems
- ✓ Anomaly Detection
- ✓ Information Retrieval

Example, in a microenterprise dataset, businesses may be grouped based on similarity in:

- ✓ Sales
- ✓ Location
- ✓ Risk level
- ✓ Years of operation

TYPES OF DATA AND DISTANCE MEASURES

THE TYPE OF DATA DETERMINES HOW WE COMPUTE SIMILARITY.

A. Numeric Data (Continuous Variables)

Examples:

- ✓ Sales revenue
- ✓ Temperature
- ✓ Profit margin

Distance Measures:

- ✓ Euclidean Distance
- ✓ Manhattan Distance

B. Binary Data (0 or 1)

Examples:

- ✓ Has insurance (1/0)
- ✓ Has backup generator (1/0)

Distance Measures:

- ✓ Simple Matching Coefficient (SMC)
- ✓ Jaccard Coefficient

C. Categorical Data

Examples:

- ✓ Business type (Retail, Food, Services)
- ✓ Location type (Urban, Rural)
- ✓ Insurance Status (Yes, No)

Distance Measures:

- ✓ Simple Matching Coefficient (SMC)
- ✓ Hamming Distance
- ✓ Jaccard Coefficient

D. Ordinal Data

Examples:

- ✓ Risk level (Low, Medium, High)
- ✓ Satisfaction rating (1–5)

Converted to ranks before computing distance.

Distance Measures:

- ✓ Rank + Numeric distance
- ✓ Euclidean/Manhattan Distance

DISTANCE VS SIMILARITY: CONCEPTUAL DIFFERENCE

Distance measures **how far apart** two objects are.

- ✓ Larger value → more different
- ✓ Smaller value → more similar
- ✓ 0 → identical

Examples:

- ✓ Euclidean distance
- ✓ Manhattan distance
- ✓ Hamming distance

Similarity measures **how alike** two objects are.

- ✓ Larger value → more similar
- ✓ Smaller value → more different
- ✓ 1 (or maximum) → identical

Examples:

- ✓ Jaccard Similarity Coefficient
- ✓ Simple Matching Coefficient

NUMERIC DATA

Example Dataset

We define two businesses:

Business A

Monthly Sales = 120 (thousand pesos)

Employees = 8

$$A = (120, 8)$$

Business B

Monthly Sales = 180 (thousand pesos)

Employees = 15

$$B = (180, 15)$$

NUMERIC DATA

1. Euclidean Distance

Used in:

- ✓ K-Means clustering
- ✓ KNN classification

Interpretation:

Smaller distance → more similar.

Formula:

$$d(A, B) = \sqrt{\sum (x_i - y_i)^2}$$

Step 1: Compute Differences

$$(120 - 180) = -60$$
$$(8 - 15) = -7$$

Step 2: Square Each Difference

$$(-60)^2 = 3600$$
$$(-7)^2 = 49$$

Step 3: Sum

$$3600 + 49 = 3649$$

Step 4: Square Root

$$\sqrt{3649} = 60.41$$

$d_{Euclidean}$ = 60.41

Final Answer:

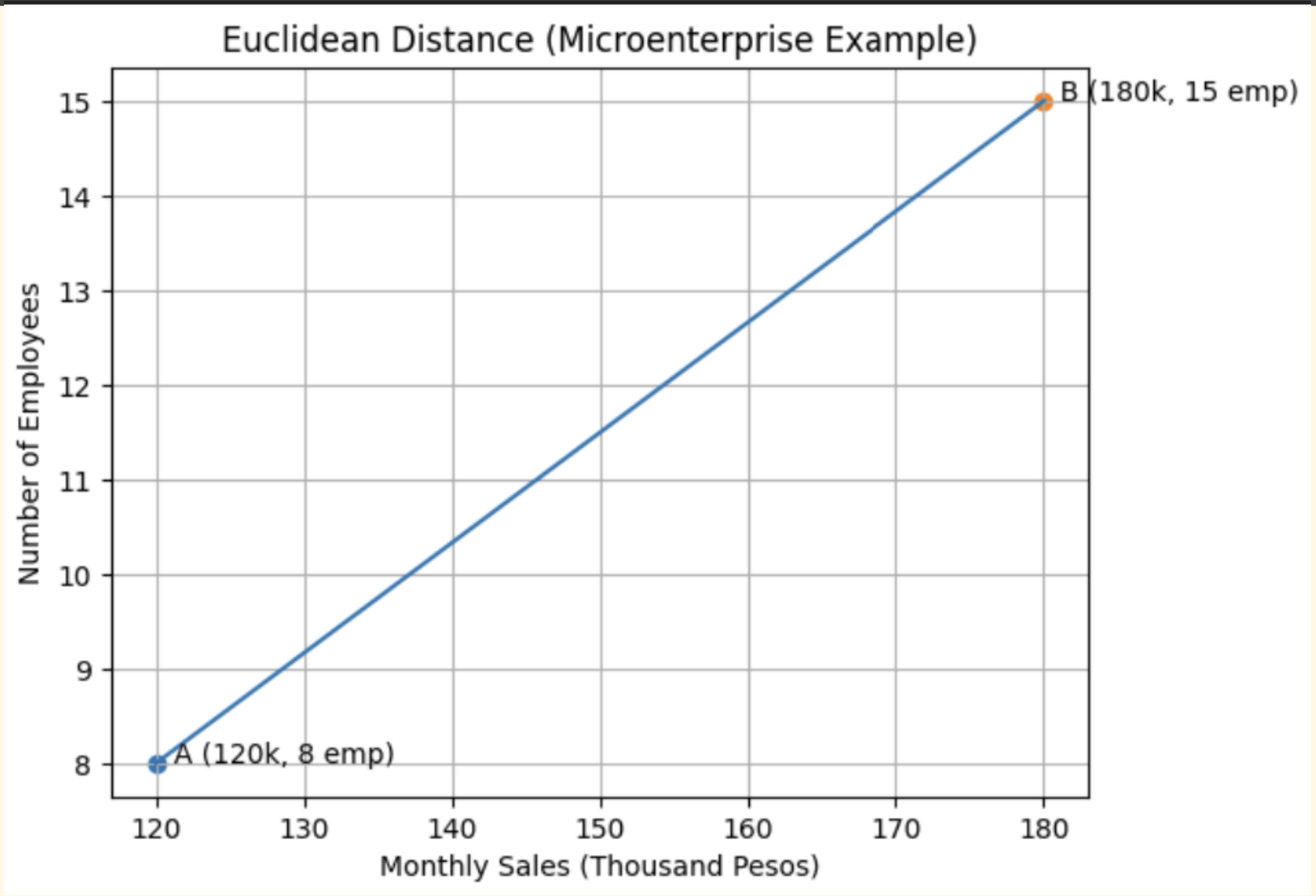
Euclidean Distance = 60.41

Interpretation:

The straight-line distance between A and B in 3D space is 60.41

NUMERIC DATA

1. Euclidean Distance



2. Manhattan Distance

Better when:

✓ Data has outliers

✓ High-dimensional datasets

Formula:

$$d(A, B) = \sum | x_i - y_i |$$

Step 1: Absolute Differences

$$| 120 - 180 | = 60$$

$$| 8 - 15 | = 7$$

Type equation here.

Step 2: Sum

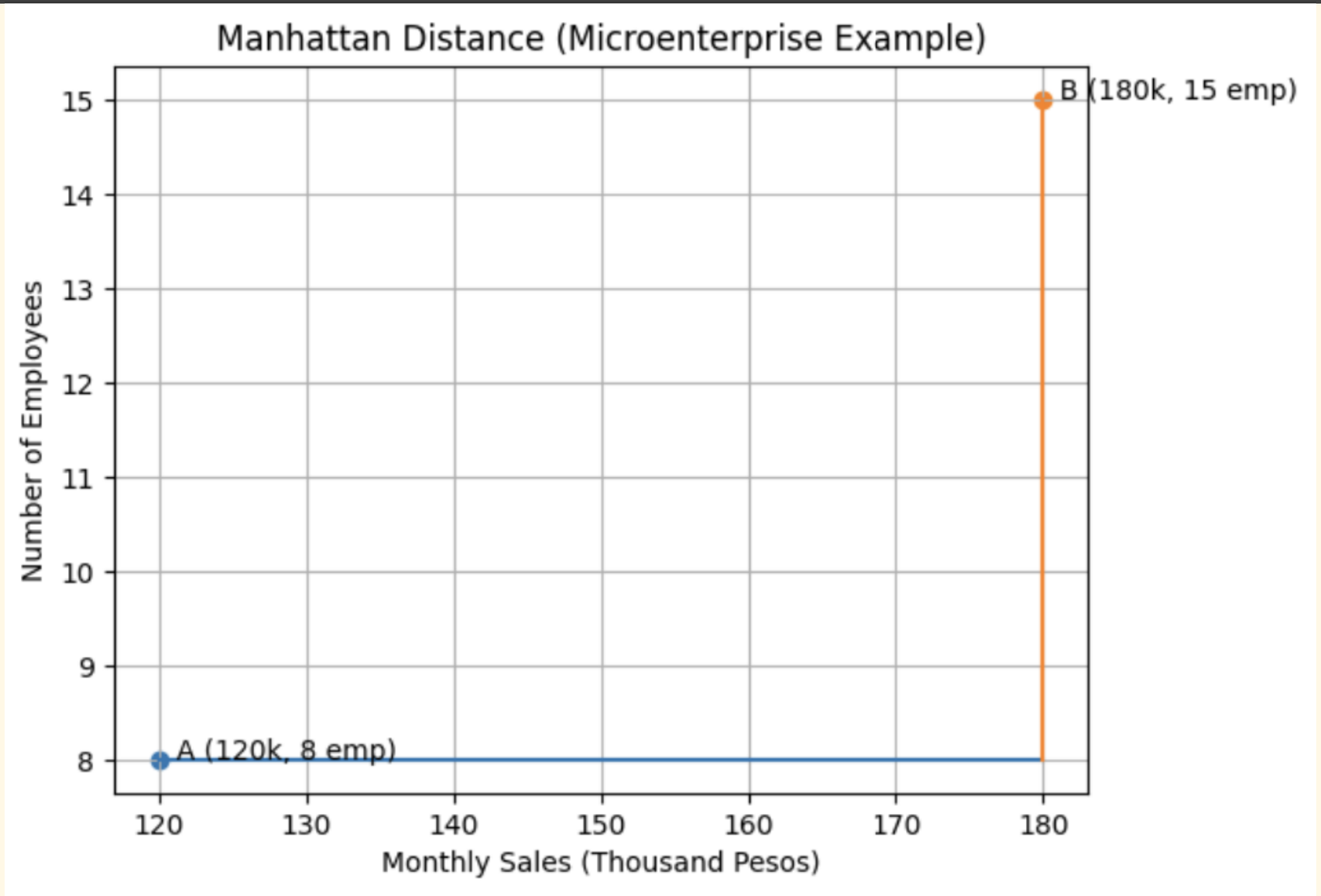
$$60 + 7 = 67$$

Final Answer: $d_{Manhattan}$ **= 67**

Interpretation:

Total grid-based distance (like city blocks) between A and B is 67.

2. Manhattan Distance



BINARY DATA (0 OR 1)

Examples:

- ✓ Has insurance (1/0)
- ✓ Has backup generator (1/0)

Distance Measures:

- ✓ Simple Matching Coefficient (SMC)
- ✓ Jaccard Coefficient

Suppose we have 4 preparedness indicators:

1. Has Generator
2. Has Insurance
3. Has Emergency Plan
4. Has Backup Data

Binary Encoding:

- ➢ 1 = Yes
- ➢ 0 = No

BINARY DATA (0 OR 1) - SMC

Feature	A	B	Result
Generator	1	1	a (1–1)
Insurance	1	0	b (1–0)
Emergency Plan	0	1	c (0–1)
Backup Data	1	1	a (1–1)

Step 1: Count carefully

- ✓ a = 2 (1–1 matches)
- ✓ b = 1 (1–0)
- ✓ c = 1 (0–1)
- ✓ d = 0 (0–0)

Total attributes:

$$a + b + c + d = 2 + 1 + 1 + 0 = 4$$

Step 2: SMC Formula

$$SMC = \frac{a + d}{a + b + c + d}$$

Substitute values:

$$SMC = \frac{2 + 0}{4}$$

$$SMC = \frac{2}{4}$$

$$SMC = 0.5$$

Compute distance:

$$\text{Distance}_{SMC} = 1 - 0.5 = 0.5$$

✅ Interpretation: 50% dissimilar, 50% similar.

BINARY DATA (0 OR 1) - SMC

Feature	A	B	Result
Generator	1	1	a
Insurance	1	0	b
Emergency Plan	0	0	d
Backup Data	1	1	a

Step 1: Count carefully

- ✓ a = 2 (1–1 matches)
- ✓ b = 1 (1–0)
- ✓ c = 0 (0–1)
- ✓ d = 1 (0–0)

Total attributes:

$$a + b + c + d = 2 + 1 + 0 + 1 = 4$$

Step 2: SMC Formula

$$SMC = \frac{a + d}{a + b + c + d}$$

Substitute values:

$$SMC = \frac{2 + 1}{4}$$

$$SMC = \frac{3}{4}$$

$$SMC = 0.75$$

Compute distance:

$$\text{Distance}_{SMC} = 1 - 0.75 = 0.25$$

✅ Interpretation: 25% dissimilar, 75% similar.

BINARY DATA (0 OR 1) – JACCARD COEFFICIENT

Feature	A	B	Result
Generator	1	1	a (1–1)
Insurance	1	0	b (1–0)
Emergency Plan	0	1	c (0–1)
Backup Data	1	1	a (1–1)

Jaccard ignores (0,0) matches. It only considers presence (1's).

Step 1: Count carefully

- ✓ a = 2 (1–1 matches)
- ✓ b = 1 (1–0)
- ✓ c = 1 (0–1)
- ✓ d = 0 (0–0)

Total attributes:

$$a + b + c = 2 + 1 + 1 = 4$$

Step 2: Jaccard Formula

$$J = \frac{a}{a + b + c}$$

Substitute values:

$$J = \frac{2}{4}$$

$$J = 0.5$$

Compute distance:

$$\text{Distance}_{Jaccard} = 1 - 0.5 = 0.5$$

✅ Interpretation: 50% dissimilar, 50% similar.

BINARY DATA (0 OR 1) - JACCARD

Feature	A	B	Result
Generator	1	1	a
Insurance	1	0	b
Emergency Plan	0	0	d
Backup Data	1	1	a

Jaccard ignores (0,0) matches. It only considers presence (1's).

Step 1: Count carefully

- ✓ a = 2 (1–1 matches)
- ✓ b = 1 (1–0)
- ✓ c = 0 (0–1)
- ✓ d = 1 (0–0)

Total attributes:

$$a + b + c = 2 + 1 + 0 = 3$$

Step 2: Jaccard Formula

$$J = \frac{a}{a + b + c}$$

Substitute values:

$$J = \frac{2}{3}$$

$$J = 0.67$$

Compute distance:

$$\text{Distance}_{SMC} = 1 - 0.67 = 0.33$$

✅ Interpretation: 33% dissimilar, 67% similar.

IMPORTANT NOTES

- This formula **only applies to similarity measures bounded between 0 and 1**, like SMC or Jaccard.
- For numeric distances (Euclidean, Manhattan), you **do not subtract from 1**. They already measure dissimilarity directly.
- This conversion is especially useful when:
  - Using similarity measures in clustering algorithms that expect distances (e.g., hierarchical clustering).
  - Mixing multiple similarity measures for comparison.

Microenterprise Categorical Dataset Example

Suppose we have **3 microenterprise attributes (categorical)**:

1. **Business Type** → {Retail, Food, Services}
2. **Location** → {Urban, Rural}
3. **Insurance Status** → {Yes, No}

CATEGORICAL DATA - SMC

Compare attribute by attribute

Feature	A	B	Match?
Type	Retail	Food	No (0)
Location	Urban	Urban	Yes (1)
Insurance	Yes	No	No (0)

$$SMC = \frac{total\ Matches}{total\ Attributes}$$

$$SMC = \frac{1}{3}$$

$$SMC = 0.33$$

Distance_{SMC}= 1−0.333 = 0.667

✅ Interpretation: 33% similar, 67% different.

CATEGORICAL DATA - HAMMING

Compare attribute by attribute

Feature	A	B	Match?
Type	Retail	Food	No (0)
Location	Urban	Urban	Yes (1)
Insurance	Yes	No	No (0)

$$\text{Hamming} = \frac{total\ Mismatches}{total\ Attributes}$$

$$\text{H}amming = \frac{2}{3}$$

$$Hamming = 0.667$$

✅ Interpretation: 2 out of 3 attributes are different.

CATEGORICAL DATA - JACCARD COEFFICIENT

Jaccard is mostly used for binary attributes, but for categorical data, we can encode each category as a binary indicator (one-hot encoding).

Feature	Category	A	B
Type	Retail	1	0
Type	Food	0	1
Type	Services	0	0
Location	Urban	1	1
Location	Rural	0	0
Insurance	Yes	1	0
Insurance	No	0	1

Now each feature-category pair is binary.

Step 1: Apply Jaccard

Jaccard similarity:

$$J = \frac{a}{a + b + c}$$

Where:

- ✓ a = # of attributes both 1 (1−1)
- ✓ b = # of 1−0
- ✓ c = # of 0−1

Count:

a = 2 → (Location:1−1, Type: both 0 for Services? Actually Jaccard ignores 0−0, so we ignore 0−0) → Only Location:1−1 → a=1

b = 2 → A has 1, B has 0 → Retail, Yes, Insurance Yes? Actually let's count:

- Type: Retail 1−0 → b=1
- Insurance Yes 1−0 → b=2

c = B has 1, A has 0 → Food:0−1, Insurance No:0−1 → c=2

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Where:

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- ✓ b = # of 1–0
- ✓ c = # of 0–1

Step 2: Compute Jaccard similarity

$$J = \frac{a}{a + b + c} = \frac{1}{1 + 2 + 2} = \frac{1}{5} = 0.2$$

Step 3: Distance

$$\text{Distance}_{Jaccard} = 1 - 0.2 = 0.8$$

✅ Interpretation: Only 20% of **positive category matches** are shared.

CATEGORICAL DATA - SUMMARY TABLE

Measure	Similarity	Distance
SMC	0.333	0.667
Hamming	N/A	0.667
Jaccard	0.2	0.8

ORDINAL DATA

Ordinal data is categorical data with a **natural order**, but the difference between levels may not be exactly equal.

Example for microenterprises:

Attribute	Scale (Ordinal)
Risk Level	Low = 1, Medium = 2, High = 3
Preparedness	Poor = 1, Average = 2, Excellent = 3
Satisfaction	Low = 1, Medium = 2, High = 3

Business Profiles

Business A

- ✓ Risk Level = Medium (2)
- ✓ Preparedness = Poor (1)
- ✓ Satisfaction = High (3)

$$A = (2,1,3)$$

Business B

- ✓ Risk Level = High (3)
- ✓ Preparedness = Average (2)
- ✓ Satisfaction = Medium (2)

$$B = (3,2,2)$$

Business Profiles

Business A

- ✓ Risk Level = Medium (2)
- ✓ Preparedness = Poor (1)
- ✓ Satisfaction = High (3)

$$A = (2,1,3)$$

Business B

- ✓ Risk Level = High (3)
- ✓ Preparedness = Average (2)
- ✓ Satisfaction = Medium (2)

$$B = (3,2,2)$$

Step 1: Normalize Ordinal Values

For some distance measures (like Euclidean), it's recommended to **normalize ordinal ranks** to [0,1]:

$$x' = \frac{x-1}{k-1}$$

Where:

- ✓ x = ordinal value
- ✓ k = number of levels

Here, k = 3 for all attributes.

Normalize A

$$A' = \left(\frac{2-1}{2}, \frac{1-1}{2}, \frac{3-1}{2}\right) = (0.5,0,1)$$

Normalize B

$$B' = \left(\frac{3-1}{2}, \frac{2-1}{2}, \frac{2-1}{2}\right) = (1,0.5,0.5)$$

Step 2: Compute Distance Measures

Euclidean Distance

$$d(A', B') = \sqrt{\sum_{i=1}^{3} (A'_i - B'_i)^2}$$

$$= \sqrt{\left(0.5 - 1\right)^2 + \left(0 - 0.5\right)^2 + \left(1 - 0.5\right)^2}$$

$$= \sqrt{\left(-0.5\right)^2 + \left(-0.5\right)^2 + \left(0.5\right)^2}$$

$$= \sqrt{0.25 + 0.25 + 0.25} = \sqrt{0.75} \approx 0.866$$

✅ Euclidean distance ≈ 0.866

Manhattan Distance

$$d(A', B') = \sum_{i=1}^{3} | A'_i - B'_i |$$

$$=| 0.5 - 1 |+| 0 - 0.5 |+| 1 - 0.5 |$$

$$= 0.5 + 0.5 + 0.5 = 1.5$$

DATA

- Attributes/Features
- Types of Data Sets
- Data Quality
- Data Preprocessing
- Similarity and Dissimilarity
- **Density**

DENSITY

Density refers to **how closely data points are packed together in the feature space**. It is a key concept in clustering, anomaly detection, and identifying dense regions of data.

- ✓ High-density regions → many similar data points clustered together
- ✓ Low-density regions → sparse, possibly outliers or unique behaviors

Density-based methods are especially useful in microenterprise data for:

- ✓ Identifying clusters of **similar businesses** based on sales, risk, or preparedness
- ✓ Detecting **outliers**, like underperforming or high-risk microenterprises

DENSITY

In data mining, **density** measures how many data points exist in a region of space.

General idea:

Density at point x:

$$\text{Density}(x) = \frac{\textit{Number of points in neighborhood of } x}{\textit{Volume of that neighborhood}}$$

The interpretation depends on:

- ✓ How we define the neighborhood
- ✓ Whether we divide by volume
- ✓ Whether we weight by distance

Example Dataset (2D Microenterprise Data)

We use two features:

- ✓ Monthly Sales (in thousands)
- ✓ Number of Employees

Business	Sales	Employees
A	50	5
B	55	6
C	52	5
D	100	15
E	48	4

We compute density around **Business A (50, 5)**.

Business	Sales	Employees
A	50	5
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We compute density around **Business A (50, 5)**.

Density Using Fixed Radius (ε-neighborhood)
Step 1: Choose radius
Let:

$$r = 6$$

Step 2: Compute Euclidean distances from A
Euclidean formula:

$$d(x, y) = \sqrt{(x_1 - y_1)^2 + (x_2 - y_2)^2}$$

Distance A → B

$$d(A, B) = \sqrt{(50 - 55)^2 + (5 - 6)^2}$$
$$= \sqrt{25 + 1}$$
$$= \sqrt{26}$$
$$= 5.10$$

✓Inside radius 6

Business	Sales	Employees
A	50	5
B	55	6
C	52	5
D	100	15
E	48	4

We compute density around **Business A (50, 5)**.

Distance A → C

$$d(A, C) = \sqrt{(50-52)^2 + (5-5)^2}$$
$$= \sqrt{4}$$
$$= 2$$

✓Inside radius

Distance A → D

$$d(A, D) = \sqrt{(50-100)^2 + (5-15)^2}$$
$$= \sqrt{2500 + 100}$$
$$= \sqrt{2600}$$
$$= 50.99$$

✕Outside radius

Business	Sales	Employees
A	50	5
B	55	6
C	52	5
D	100	15
E	48	4

We compute density around **Business A (50, 5)**.

Distance A → E

$$d(A, E) = \sqrt{(50-48)^2 + (5-4)^2}$$
$$= \sqrt{4+1}$$
$$= \sqrt{5}$$
$$= 2.24$$

✓Inside radius

Step 3: Count points inside radius
Inside radius:
A
B
C
E
Total = 4 points

DENSITY

Business	Sales	Employees
A	50	5
B	55	6
C	52	5
D	100	15
E	48	4

Step 4: Compute Area (2D case)
Neighborhood is a **circle**:

$$\begin{aligned}\text{Area} &= \pi r^2 \\ &= \pi\,(6)^2 \\ &= 36\pi \\ &\approx 113.10\end{aligned}$$

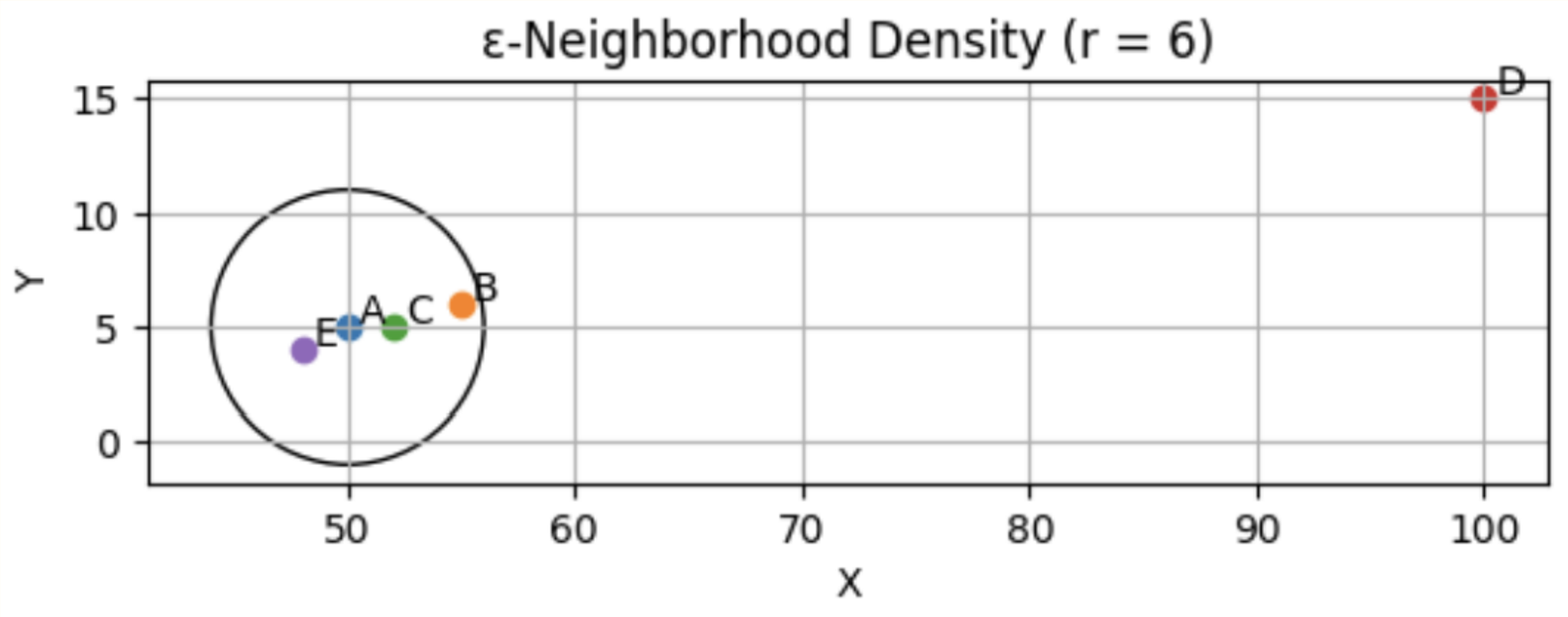
Step 5: Compute Density

$$\text{Density}(x) = \frac{\textit{Number of points in neighborhood of } x}{\textit{Volume of that neighborhood}}$$

$$\begin{aligned}\text{Density}(A) &= \frac{4}{113.10} \\ &= 0.0354\end{aligned}$$

DENSITY

Business	Sales	Employees
A	50	5
B	55	6
C	52	5
D	100	15
E	48	4



POST ASSESSMENT

- ✓ DM-DM4-DistanceMeasure-Sy.ipynb
- ✓ DM-ME5-Density-Sy.ipynb

THANK YOU!

ANY QUESTIONS?